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[ANUVUTI: A Video Dataset for Emotion Recognition in Bangla Languages]

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Keywords

Emotion recognition; video-based emotion analysis; low-resource languages; Bangla speech variations; machine learning applications; multimodal datasets; affective computing; human-computer interaction.

Abstract

The ANUVUTI dataset is a comprehensive collection designed for video-based emotional expression analysis in Bangla, addressing the scarcity of datasets for emotion recognition in low-resource languages. This dataset fills a gap by providing essential resources for Bangla-based emotion recognition research. Data were collected over one month using Smartphone's, capturing performances from 10 non-professional actors (6 female, 4 male) aged 18 to 25. The actors portrayed seven emotions: happiness, sadness, anger, fear, disgust, surprise, and neutrality. The total recordings include 5 sentences × 3 repetitions × 6 emotions (anger, happiness, sadness, fear, disgust, and surprise) × 10 actors, amounting to 900 recordings, plus 5 sentences × 3 repetitions × 1 emotion (neutral) × 9 actors, resulting in 135 additional recordings. This brings the dataset to a total of 1,035 video clips. Informed consent was obtained from all participants, ensuring transparency regarding the dataset's purpose and potential uses. Actors incorporated regional Bangla speech variations, enhancing the authenticity of emotional expressions. The final dataset, processed to maintain consistent 720p resolution at 25 frames per second, has a total size of 497 MB. The ANUVUTI dataset offers extensive reuse potential for developing and testing machine learning models for emotion-aware systems, such as virtual assistants, sentiment analysis tools, and adaptive interfaces. It also serves as a benchmark for multimodal emotion recognition and speech-to-emotion applications. Furthermore, the dataset facilitates interdisciplinary research by integrating insights from computer science, linguistics, psychology, and human-computer interaction, advancing the understanding and application of emotion recognition technologies.

SPECIFICATIONS TABLE

[Instruction for SPECIFICATIONS TABLE in comment box]

Subject	Computer Science , Multimedia, Data Science and Human-Computer Interaction (HCI)
Specific subject area	Video-based emotion detection dataset focusing on the Bangla language.
Type of data	Raw Video (mp4)
Data collection	The ANUVUTI dataset was collected over one month using Smartphone's to capture real-time emotional expressions. Ten non-professional actors (6 female, 4 male), aged 18-25, portrayed seven emotions: happy, sad, anger, fear, disgust, surprise, and neutral. Each actor performed five Bengali sentences per emotion, repeated three times for variability, resulting in 1,035 clips. Participants provided informed consent, and privacy was maintained through signed forms. Videos were processed using MiniTool MovieMaker to ensure 720p resolution at 25fps, with regional Bengali variations for authenticity.
Data source location	East West University Bangladesh, Jahurul Islam Ave, Dhaka 1212, Bangladesh (Latitude: 23.7983, Longitude 90.3749)
Data accessibility	Repository name: Mendeley Data Data identification number:*(or DOI or persistent identifier)* Direct URL to data:… Instructionsfor accessing these data: …
Related research article	None.

VALUE OF THE DATA

- The ANUVUTI dataset offers a specialized collection of video-based emotional expressions in Bangla, a low-resource language, providing significant value for advancing emotion detection technologies and cultural studies within affective computing.
- This dataset can support the development of emotion-aware systems, such as virtual assistants and adaptive interfaces, which recognize emotions from video inputs, enhancing user experience.
- By capturing emotional expressions tied to Bangla sentences, this dataset allows researchers to work on sentiment analysis, multimodal emotion recognition, and speech-to-emotion applications.
- The use of regional Bangla speech variations adds value for studying how emotions are expressed across linguistic and cultural contexts, contributing to sociolinguistic research.
- This dataset supports research at the intersection of computer science, psychology, and linguistics, opening new avenues for emotion recognition systems used in mental health monitoring and behavioral analysis.

BACKGROUND

The ANUVUTI dataset was developed to address the scarcity of video-based emotion recognition resources tailored to the Bangla language, a gap often neglected in existing research. To enhance the performance of emotion recognition models, the dataset captures authentic emotional expressions that align with the conceptual nature of each emotion. Research indicates that non-professional actors tend to exhibit more realistic emotions compared to trained actors, who may exaggerate their expressions [11]. Consequently, ANUVUTI comprises 10 non-professional actors, aged 18 to 25, performing 35 words across seven emotional categories: happiness, anger, sadness, surprise, fear, disgust, and neutral [12] [13].

The design reflects the linguistic and cultural nuances of Bangla, providing a robust foundation for future research into emotion recognition. ANUVUTI's structured approach bridges the interaction between language, culture, and emotional expression. Additionally, the EmoV-DB framework supports video content labeling and prediction, contributing to consistent annotation practices [14]. This dataset offers valuable insights for advancing emotion recognition technologies, helping researchers build context-aware models suited to Bengali-speaking communities.

DATA DESCRIPTION

The ANUVUTI dataset is a multimodal video resource created to address the absence of Bengali-language datasets for emotion recognition. The developed ANUVUTI dataset is a realistic Bangla emotional speech dataset and Table 1 provides a descriptive summary of the dataset.

It contains 1,035 video clips across seven emotional categories: happiness, sadness, anger, fear, disgust, surprise, and neutrality. This dataset captures natural emotional expressions using non-professional actors, focusing on real-world delivery through facial expressions, speech, and audio

cues. The recordings were made using Smartphone's to ensure authenticity and accessibility. Each of the six primary emotions—happiness, sadness, anger, fear, disgust, and surprise—was expressed by 10 actors, each performing five distinct sentences. Every sentence was repeated three times, generating 150 clips for each emotion. This structure produces a total of 900 video clips for the six primary emotions, calculated as 5 sentences × 3 repetitions × 10 actors = 150 clips per emotion. For the neutral emotion, nine actors participated, following the same structure, resulting in 135 clips (5 sentences × 3 repetitions × 9 actors = 135 clips). Combining these, the total dataset consists of 900 + 135 = 1,035 video clips. The videos were recorded in various local environments, including East West University, to ensure linguistic diversity. Each clip is between 1 to 4 seconds in duration and recorded at a resolution of 720p (1280 × 720) with a frame rate of 25 frames per second (fps). All files are in MP4 format, with a sampling rate of 44.1 kHz, and the entire dataset size is approximately 497 MB. The clips are organized into seven folders based on the emotion category, with clear naming conventions. Each file name in the ANUVUTI dataset follows the specific structure:[ActorID]_[Emotion][SentenceNumber]_[Repetition].mp4.This format ensures that each file uniquely identifies the actor, the emotion being expressed, the specific sentence used, and the repetition of the performance. The file name (e.g. a video named Actor01_Neutral_04_2.mp4) represents the first actor expressing the fourth sentence for the neutral emotion in its second repetition, while a file labeled Actor03_Happy_01_1.mp4 contains the third actor performing the first happiness sentence in its first repetition. This consistent naming convention makes it easy to locate specific clips, facilitating structured data handling. For example, the sentence "আমি বিকালে বের হবো" (Ami bikale ber hobo), which translates to "I will go out in the afternoon," is one of the neutral sentences included in the dataset. Phonemic transcriptions accompany the sentences to help ensure pronunciation consistency across actors, adding value to the dataset's use in multimodal emotion recognition research. Figure 1 displays the number of video clips across each emotion category within the ANUVUTI dataset.

Table 1

A summary of the ANUVUTI dataset:

Attribute	Description
Language	Standard Bangla
Emotion Categories	Happiness, Sadness, Anger, Fear, Disgust, Surprise, Neutral
Number of Clips	1,035
Clip Duration	1-4 seconds per clip
Actors	10 (6 Female, 4 Male)
Age Group	18-25 years
Resolution	720p (1280 × 720) @ 25 fps
Sampling Rate	44.1 kHz
Total Dataset Size	497 MB
Recording Location	East West University and local settings

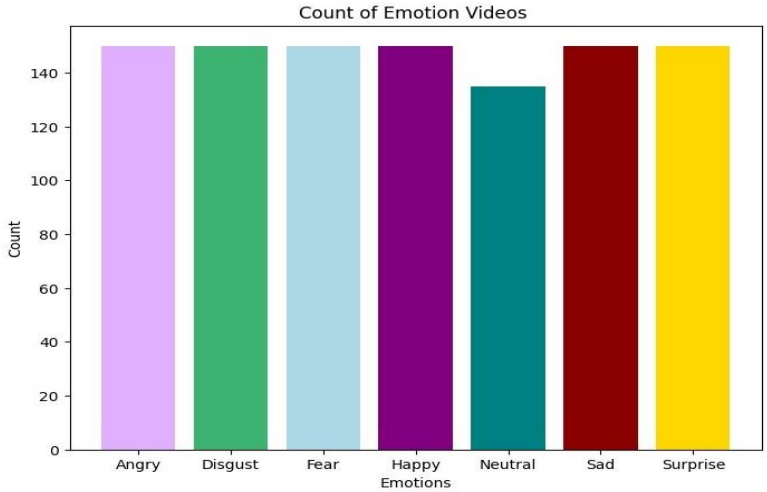

Figure 1: Count of emotion samples in the ANUVUTI dataset

It provides a clear overview of the distribution between the six primary emotions—happiness, sadness, anger, fear, disgust, and surprise—and the neutral category. With 150 clips allocated to each primary emotion and 135 clips for neutrality, it offers a quick understanding of the dataset's composition, helping researchers assess its structure briefly. Including the neutral category is crucial for establishing a baseline; ensuring models can distinguish between emotional and non-emotional states, reducing bias, and improving real-world applicability. A histogram titled Figure 2 visualizes the distribution of video lengths across all clips, confirming that most fall within the 1 to 4-second range. Additionally, to Figure 3 presents the gender distribution of participants using a pie chart, showing a 60% female and 40% male actor involvement.

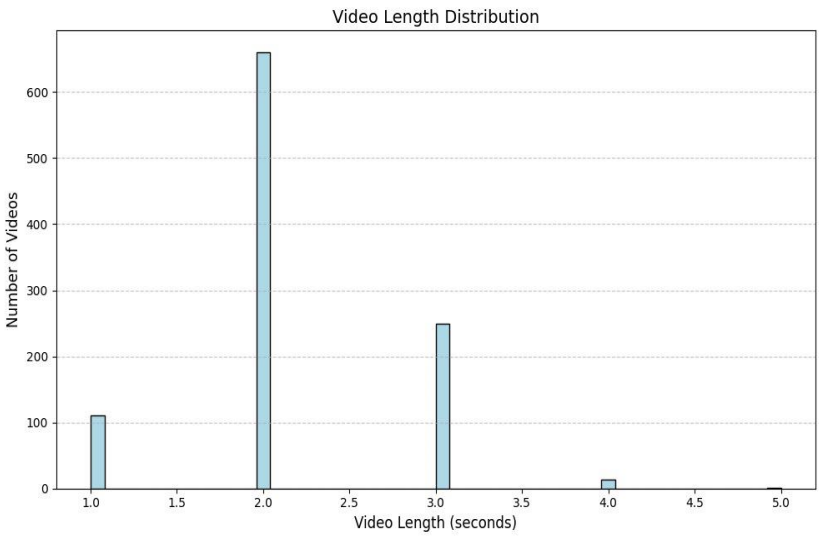

Figure 2: Histogram showing the distribution of video lengths across all clips

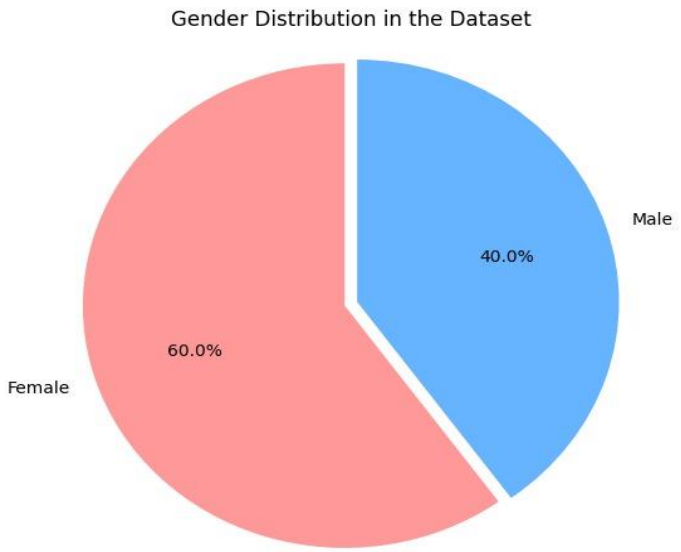

Figure 3: Pie chart showing gender distribution among participants

The VEATIC dataset [1] offers continuous valence-arousal ratings collected from videos in natural environments. Designed for dynamic emotion tracking, it emphasizes how emotions vary over time, helping to capture fluid, in-the-wild emotional states. Its real-world settings make it an essential tool for studying how emotions unfold naturally, contributing to research on affective computing. The RAVDESS dataset [2], on the other hand, provides a structured, controlled collection of emotional speech and song recordings by professional actors. It features seven core emotions at varying intensities, available in multiple modalities, including audio-only, video-only, and audio-visual, making it ideal for emotion recognition in constrained settings. MELD [4] focuses on multi-party conversations extracted from the TV show *Friends*, allowing for an analysis of emotional dynamics across interactions, blending verbal and non-verbal signals. The AFEW dataset [5] contains data drawn from films, targeting real-world conditions through facial expression analysis in noisy environments. Its role as a benchmark dataset for the EmotiW competition has established it as a critical resource for evaluating models under challenging conditions. CAER [3] takes a unique approach by combining facial expressions with contextual environmental information, aiming to understand how external factors influence emotions. This makes it an essential dataset for research into context-aware emotion recognition. The IEMOCAP dataset [6] offers a detailed exploration of multimodal emotions, combining speech and non-verbal cues in dyadic sessions. With nine emotional categories and dimensional annotations (valence, arousal, dominance), it provides 12 hours of both scripted and improvised interactions, facilitating research into complex human communication.

The AVCAffe dataset [7] addresses a modern challenge by capturing emotional and cognitive load in remote work environments. Its 58,000 video clips, collected through simulated video-conferencing scenarios, offer a unique opportunity to study how affect and cognitive load affect remote work dynamics**.** SEMAINE [8] dataset is a study on emotion recognition in virtual agents, capturing spontaneous conversations between humans and four avatar personalities. The dataset provides valuable data for studying human-agent communication and understanding emotions and cognition. It is essential for advancing human-computer interaction and affective computing by developing

models for dynamic, real-time interactions between humans and artificial agents. Collectively, these datasets offer a broad range of emotional contexts, modalities, and linguistic settings; however, they lack representation for languages like Bangla, which is addressed by ANUVUTI. ANUVUTI stands out by offering an audio-visual dataset specifically in Bangla, capturing seven basic emotions with high-resolution video and audio, thereby advancing research in localized emotion recognition. Below, Table 2 presents a detailed comparison of ANUVUTI with the prominent existing datasets.

Table 2

A Comparative Summary among Different Public Datasets with the Developed ANUVUTI Dataset

Dataset	Language	Modality	Number of Emotion	Number of Clips	Actors	Age Group	Clip Duration	Resolution	Total Dataset Size	Sampling Rate
VEATIC [1]	English	Video	8 (Various emotional dimensions)	124	192 (varied gender representation)	Varied	10s - 2min 37s	1920x1080	8.1 GB	N/A
RAVDESS [2]	English	Audio-visual	8 (neutral, calm, happy, sad, angry, fearful, surprise, and disgust)	7,356	24 (12 Female, 12 Male)	no specific age group specified	Varied (5s - 20s)	1080p (1920 × 1080)	24.8 GB	48 kHz
CAER [3]	English	Audio-visual	7(anger, disgust, fear, happy, sad, surprise, neutral	13,201	6 (varied gender representation)	Varied	1s - 5s	720p (1280 × 720)	2.5 GB	44.1 kHz
MELD [4]	English	Audio-visual	7 (Anger, Disgust, Fear, Joy, Neutral, Sadness, Surprise)	1,433	3 (varied gender representation)	Varied (multiple age groups)	3.59s	720p (1280 × 720)	3.2 GB	48 kHz
AFEW [5]	English	Video	7(anger, disgust, fear, happiness, sadness, surprise, and neutral)	1,809	3(varied gender representation	Varied (mostly adults)	0.3s - 5.4s	720p (1280 × 720)	1.5 GB	N/A
IEMOCAP [6]	English	Audio-visual	9 (angry, excited, fear, sad, surprised, frustrated, happy, disappointed, neutral)	302	10 (varied gender representation)	Varied	Varied	N/A	12 hours	48 kHz
AVCAffe [7]	English	Audio-visual	2 (Cognitive Load, Affect)	58,000	106(balanced male-female)	18-57 years	Varied	N/A	108 hours	44.1 kHz
SEMAINE [8]	English	Audio-visual	5 Epistemic States(agreement, interested, certain, concentration, thoughtful)	190	150	Varied	5 min	352 × 288 (25 fps)	N/A	N/A
Developed ANUVUTI	**Bangla**	**Video**	**7 (Happiness, Sadness, Anger, Fear, Disgust, Surprise, Neutral)**	**1,035**	**10 (6 Female, 4 Male)**	**18-25 years**	**1-4 seconds**	**720p (1280 × 720)**	**497 MB**	**44.1 kHz**

Data Sampling

To develop the ANUVUTI dataset, we focused on creating a multimodal corpus featuring both audio and visual modalities for emotion recognition in Bangla. Our dataset includes seven key emotions: happiness, sadness, anger, fear, surprise, disgust, and neutral. For data collection, we enlisted 10 speakers—4 male and 6 female—who performed emotional expressions by acting out 35 Bangla sentences specifically crafted for this purpose. Each emotional expression was captured through video recordings, ensuring high-quality visual and audio data for precise analysis. The sampling process was carefully designed to maintain ethical standards. Participants provided informed consent through a Participant Consent Form before contributing to the dataset. Video recordings were organized into seven folders, one for each emotional category, facilitating the training and evaluation of multimodal recognition systems. The recordings were balanced across gender, with consistent representation of all emotions to avoid data bias.

These samples allow the deep learning models to effectively train for emotion detection, combining facial expressions and speech data. Below Figure 4 is a statistical breakdown of the collected data samples by class for the ANUVUTI dataset. These samples enable deep learning models to train effectively, leveraging both facial expressions and speech data. The structured class distribution ensures balanced representation, which is essential for robust emotion recognition across diverse emotional categories.

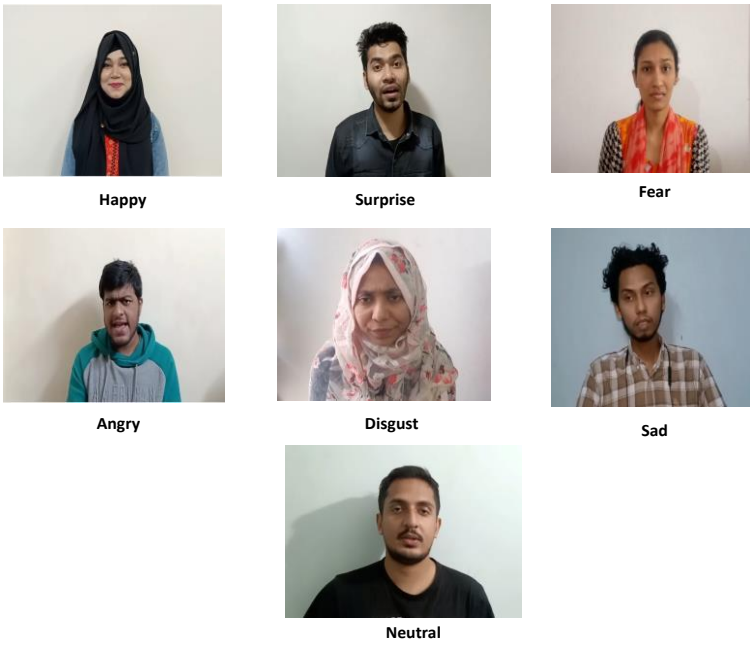

Figure 4: Sample of 7 Types of Unique Emotion Image Samples

From the ANUVUTI Dataset

Structure of the Dataset Repository

Our dataset repository is organized into a root folder containing seven subfolders, each representing a specific emotional category: Happiness, Sadness, Anger, Fear, Disgust, Surprise, and Neutral. Each folder stores the video clips corresponding to one of these emotions, facilitating structured data management. The six primary emotions—Happiness, Sadness, Anger, Fear, Disgust, and Surprise—each contain 150 clips, recorded by 10 actors. Every actor performed five different sentences for each emotion, repeating each sentence three times, resulting in 150 videos per emotion (5 sentences × 3 repetitions × 10 actors). The Neutral folder holds 135 clips, collected from nine actors following the same structure, yielding a total of 1,035 videos across the entire dataset.

We collected the data in various local environments, including East West University, to ensure authenticity and reflect real-world settings. All videos are stored in MP4 format at 720p resolution (1280 × 720) with a frame rate of 25 fps, ensuring consistency in quality. Additionally, each file name follows a specific naming pattern:[ActorID]_[Emotion][SentenceNumber]_[Repetition].mp4. For example, “Actor01_Happy_01_1.mp4” refers to the first actor expressing the first happiness sentence in the first repetition, while “Actor02_Neutral_04_3.mp4” indicates the second actor performing the fourth neutral sentence in the third repetition.

This structured organization helps ensure easy navigation and efficient data handling. The inclusion of multiple recording locations also adds linguistic diversity, enhancing the dataset’s relevance for Bangla emotion recognition research. Below is a conceptual overview of the folder hierarchy, showcasing how we organized the data for smooth accessibility and usability in multimodal recognition tasks.

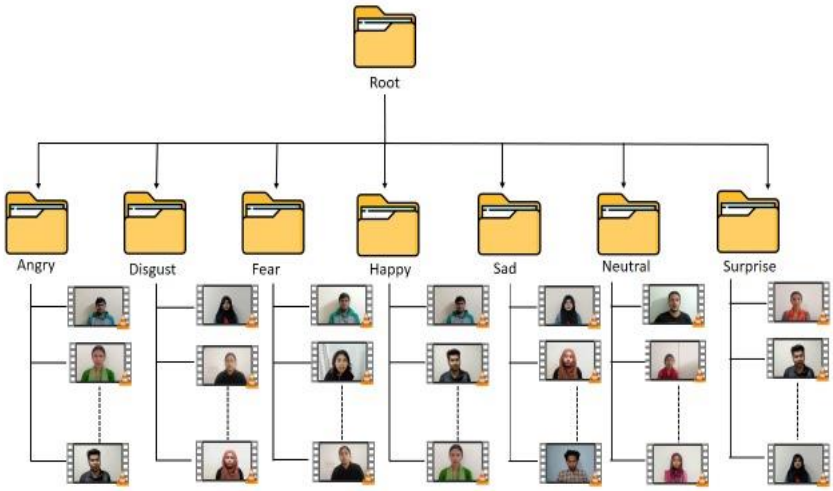

Figure 5: ANUVUTI Dataset Directory Structure

EXPERIMENTAL DESIGN, MATERIALS AND METHODS

We outline how we conduct the study of developing a systematic approach to collect, annotate, and evaluate a diverse video dataset for emotion detection, including detailed information about the processes used for participant selection, data acquisition, annotation processes, data preparation, and preprocessing. However, we provide below a detailed step-by-step protocol to describe our systematic approach to developing the ANUVUTI dataset in Fig 1. We also describe a detailed description of each step of our framework in the following subsections.

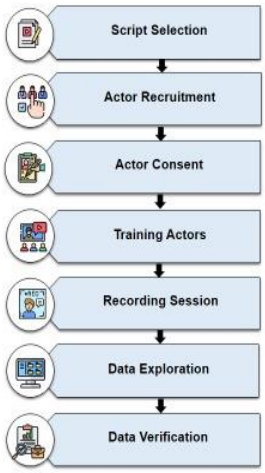

Figure 6: Workflow diagram of the ANUVUTI dataset

4.1 Selection of emotions: We chose seven emotion states – Angry, Disgust, Sad, Happy, Fear, Surprise, and Neutral. We used these seven emotions as the basic ones for people because they are all recognized by a lot of cross-cultural work and understanding in emotion science. Research such as the work of widely accepted psychological theory provided by Paul Ekman has shown “six basic emotions”[9] (Happy, Sad, Angry, Fear, Surprise, and Disgust) that these are universal expressions which means people around the world express then through similar movements. For example, anger and fear emotion state provide negative emotion recognition, and surprise and happiness allow us to detect more positive and unexpected emotional states [9]. We have also introduced a neutral baseline or control emotion for the quiet, non-emotional condition that makes distinguishing emotions and expressions easier. We included neutral as it is necessary to ensure the accurate representation of a true neutral state in datasets [10].

4.2 Script Selection: We selected 35 sentences, and divided them into seven emotion states: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise. Each state contains five emotionally relevant sentences to ensure the identification of desired emotions in video data based on their alignment with the intended emotional context. This alignment is essential for accurate emotional expression and ensures facial and speech expressions closely match the expected emotional state. Since the actors are non-professionals, it is

important to guide them so that they can authentically express emotions throughout their recordings. These sentences are simple and design to express a specific emotion state. It helps actor to memorize properly. For example, the sentence “উফ! কী বাজে দূর্গন্ধ!” (English: "Oops! What a bad smell!") Effectively convey the disgust emotion naturally. Selecting sentences based on the specific emotion was necessary because by aligning the sentences closely with emotions, non-professional actors could deliver emotions when performing them on camera in a way that those expressions would express realistically. The sentences are given below with English and phonemic translation:

Emotions	**Sentences**
Angry	বাংলা: আমারসাথেএকদমচিল্লাবানা। English: Don't mess with me at all. Phonemic: Amar satheekdomcillabana.
	বাংলা: তুমিএইব্যাপারেআমারথেকেবেশিজানো? English: You know more about this than me? Phonemic: tumieibzapareamarthekebesijano?
	বাংলা: এখানথেকেযাওতুমি। English: You go from here. Phonemic: ekhanthekezautumi
	বাংলা: এতবিরক্তকরবানাতো। English: Don't be so annoying. Phonemic: Eto biroktokorbana to.
	বাংলা: মুখসামলেকথাবলবে। English: Speak with mouth control. Phonemic: Mukh samle kotha bolbe.
Disgust	বাংলা: উফ! কীবাজেদূর্গন্ধ! English: Oops! What a bad smell! Phonemic: uff! Ki bajedurgondho!
	বাংলা: উফ! তুমিএইকাজটিকেনকরলে? English: Oops! Why did you do this? Phonemic: Uff! tumieikajti keno korle?
	বাংলা: আপনিআমারদিকেএভাবেতাকিয়েআছেনকেন? English: Why are you looking at me like this? Phonemic: Aponi amar dike evabetakiyeachen keno?
	বাংলা: উফ! সেখুবইবিরক্তিকর। English: Whoops! He is very annoying. Phonemic: Uff! Se khubibiroktikor.
	বাংলা: উফ! তাকেদেখলেইআমারগাজ্বালাকরে। English: Whoops! Just seeing him makes my heart burn. Phonemic: Uff! takedekhleiamar ga jala kore.
Fear	বাংলা: আমারঅনেকভয়লাগছে। English: I feel very scared. Phonemic: Amar onekvoylagche
	বাংলা: কে! কে! কে! ওখানে।

	English: Who! Who! Who! there. Phonemic: ke!ke!ke! Okhane
	বাংলা: আগামীকালেরপরীক্ষানিয়েআমিখুবইভয়েআছি। English: I am very scared about tomorrow's exam. Phonemic: Agamikalerporikkhaniyeamikhuboivoyeachi.
	বাংলা: সাপটিআমারদিকেআসছে! English: The snake is coming towards me! Phonemic: Saptiamar dike asche.
	বাংলা: হররমুভিটিখুবইভয়ংকরছিল। English: The horror movie was very scary. Phonemic: Horor muvitikhuboivayongkorchilo
Happy	বাংলা: আজকেআমিঅনেকখুশি। English: I am very happy today. Phonemic: ajkeamionekkhusi.
	বাংলা: তোমারসাথেদেখাহওয়াতেঅনেকভালোলাগছে। English: Nice to meet you. Phonemic: tomarsathedekhahoyateonekvalolagche.
	বাংলা: আজআমারচাকুরীহয়েছে। English: I got my job today. Phonemic: ajamarcakurihoyeche.
	বাংলা: আমিপরিক্ষায়খুবভালোকরেছি। English: I did very well in the exam. Phonemic: amiporikkhaykhubvalokorechi.
	বাংলা: বাহ! আমিতোমারসফলতায়অনেকখুশিহয়েছি। English: Wow! I am very happy with your success. Phonemic: bah! Ami tOmarsofolotayonekkhushihoyechi.
Sad	বাংলা: তারমৃত্যুআমিকল্পনাওকরতেপারিনি। English: I could not imagine his death. Phonemic: tar mrritzuamikolponaokorteparini.
	বাংলা: ইস! দূর্ঘটনাটিখুবইমর্মান্তিক। English: Yes! The accident is very tragic. Phonemic: ishh! durrghtonatikhuboimorrmantik.
	বাংলা: তোমারসাথেএটিখুবইখারাপহয়েছে। English: It's too bad for you. Phonemic: tomarsatheetikhubikharaphoyeche.
	বাংলা: আজকেরদিনটিআমারজন্যভালোযায়নি। English: Today has not gone well for me. Phonemic: Ajkerdintiamarjonnovalo jay ni
	বাংলা: কখনওকখনওভাগ্যআমাদেরপক্ষেথাকেনা। English: Sometimes luck is not on our side. Phonemic: Kokhonokokhonovaggoamderpokkhethakena
Surprise	বাংলা: আপনিফিরেএসেছেন!

	English: You are back! Phonemic: Apnoi fire esechen!
	বাংলা: উপহারটিআমারজন্য! English: The gift is for me! Phonemic: upohartiamarjonno?
	বাংলা: বাহ! কীসুন্দরএকটিজায়গা। English: Wow! What a beautiful place. Phonemic: Bah! ki sundorektijayga!
	বাংলা: তারআচরনেঅবাকহয়েছি। English: I am surprised by his behavior. Phonemic: Ami tar achoroneovakhoyechi
	বাংলা: বাহ! তুমিএইকাজটিএততাড়াতাড়িকিভাবেকরলে? English: Wow! How did you do this so quickly? Phonemic: Bah! tumieikajtietotaratarikivabekorle?
Neutral	বাংলা: আমিভালোআছি। English: I am fine. Phonemic: Ami valoachi
	বাংলা: আমিআজকেযেতেপারবনা। English: I can't go today. Phonemic: Ami ajkezeteparboń'a
	বাংলা: আপনিএইকাজটিকরতেপারবেন? English: Can you do this? Phonemic: Apnoieikajtikorteparben?
	বাংলা: আমিবিকালেবেরহবো। English: I will go out in the afternoon. Phonemic: Ami bikaleber hobo
	বাংলা: আমিগল্পেরবইপড়তেপছন্দকরি। English: I like to read storybooks. Phonemic: Ami golperboiportepochondhokori.

4.3 Actor Recruitment: Ten non-professional actors from East West University of Bangladesh (age groups between 18 and 25) Livings in Dhaka, Bangladesh, were selected randomly to portray the seven human emotional states: anger, disgust, fear, neutral, sad, surprised, and happy throughout the video recordings. We selected non-professional actors to ensure a natural representation of emotional expressions in the Bengali language. All of the actors participated in seven emotional video recordings. Among these actors, six were female, and 4 were male. We mainly selected these individuals who were representative of everyday actors so that we could ensure the dataset reflected real-world emotional expressions.

4.4 Actor Consent: We provided the information of concern form to all the actors before their video recordings. We collected consent to ensure that the actors fully understood the purpose of the study. We also explained our project objectives specifically collection of emotional expression, about the recording process, the types of emotions to portray, how we stored their data to create a trusting environment. All actors filled out their personal information and signed the concern form to proceed next steps.

4.5 Training the Actors: Since all the actors are involving were non-professional, we need to prepare the recording set suitable for them and express them our process clearly before their video recordings. Before the recordings, we introduce the actors to the seven emotions: happy, sad, angry, fear, surprise, disgust, and neutral. Firstly, we introduced a description of each emotion script and guide them to portray each emotion properly. To ensure the understanding of what emotions are recorded, actors could take time to mentally prepare and practice each emotion individually before each recording. We also arranged trial session so that we could guide them to experiment with their facial expressions, body language and vocal expression clearly to match the emotion's intensity. We encouraged them to give their personal experiences and allow them to adapt pre-selected sentences of their local or regional variations.

4.6 Recording Session: We set our recording environment to help the actors feel comfortable and relaxed, reducing unnecessary behaviour and less camera awareness during recording. We recorded videos using a Smartphone. This flexibility proved helpful in getting our non-professional actors to communicate the proper emotion management by allowing them to use words and phrases that felt more natural and familiar to them to reach target emotions. Actors could perform more freely and expressively since the sentences were adjusted for how they usually speak to give emotions that truly felt relatable. During each video recording, we informed them what emotion was recorded now and gave them time to prepare themselves. Actors faced difficulties expressing certain emotions, especially fear and sadness. Multiple takes are taken during these emotional recordings. We requested the actor to repeat each emotion sentence three times.

4.7 Data Exploration: The collection of video recordings lasted up to 1–4 s, depending on the speed at which a speaker was talking and how sentences were formulated in different emotional conditions. All videos were shot on a Smartphone, creating a natural and consistent environment across the sessions. The final dataset contains 1035 recordings from 10 participants portraying the seven human emotional states: anger, disgust, fear, neutral, sad, surprise, and happiness. These are included in this dataset for 35 sentences (we used

five sentences for each emotional state) in a realistic sense. The total number of recordings consists of 5 sentences × 3 repetitions × 6 emotional states (angry, happy, sad, fear, disgust, and surprise) × 10 participating speakers = 900 video recordings + 5 sentences × 3 repetitions × 1 emotional state (neutral) × 9 participating speakers = 135 recordings, resulting in a total of 1035 recordings.

4.7.1 Data trimming and Resolution Adjustment: In video recordings, there were instances when a single video exceeded the intended duration. This was often because of pauses or extra-lengths the actors may take, which were more than necessary for that exact sentence. That's why we trimmed videos to match according to sentence duration. We used MiniTool MovieMaker software, and the only important part was keeping the final clips to avoid some extra amendments on the video clip. These videos were then down-sampled to a vertical resolution of 720p, striking between visibility and size. All videos were done at a frame rate of 25fps for fluid motion and uniformity across all samples.

4.7.2 Filename Convention: Each actor was given a specific sequence number within our dataset at random, and the sentences were written down in advanced order of appearance. This is translated into a file naming convention allowing easily locating and organizing the recording for further processing or analysis. The detailed design of the filename structure: P_<person-number><1st-emotion><sentence-number>_ (<number-of-repeat>). For example, file name is:P_1_A_1 (1).mp4 where P_01: Refers to the first actor, A refers to the "angry" emotion, A_1 refers to the first sentence in the Angry emotion, (1): indicates that this is the first repetition of the 1st sentence by the actor.

Table 2.7.2: Description of Filename Structure of ANUVUTI dataset.

Identifier	Meaning
Actor Number	P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9, P_10
Emotion State	A=Angry, D=Disgust, F=Fear, H=Happy, N=Neutral, S=Sad, Sur=Surprise
Sentence Number	1,2,3,4,5,……35
Sentence Repetition	1(1),1(2),1(3),2(1),2(2)……..5(3)

LIMITATIONS

The ANUVUTI dataset, though valuable, has some limitations. With 1,035 videos and a narrow demographic of non-professional actors aged 18 to 25, it lacks diversity across age groups and socio-economic backgrounds. This limited variety may introduce bias and affect the generalizability of models trained on it. Additionally, cultural differences in emotional expression may lead to subjective

labelling inconsistencies. Environmental noise from real-world recording settings, though minimized, may have impacted audio or video clarity. Lastly, the dataset's size, while sufficient for initial research, might limit scalability for more complex deep learning models. Future expansions can address these challenges to enhance usability

ETHICS STATEMENT

Data in Brief's Guide for Authors contains detailed information on the ethical guidelines with which all authors must comply. In addition, we ask you to complete the relevant statement(s) below. Please delete those which are not relevant for your data.

If your work involved human subjects, *please include a statement here confirming that the relevant informed consent was obtained from those subjects, if the research was carried out in accordance with the Declaration of Helsinki, and if it includes the* ***Ethical committee approval and the protocol number****. A copy of the consent form (in English) can be submitted either in a data repository or under the section 'Consent form' while submitting the manuscript to the system.*

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